

Introduction

Session 01

Research Design and Methods in Quantitative Research

Dr. K. Jonathan Klüser

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Introduction

About me

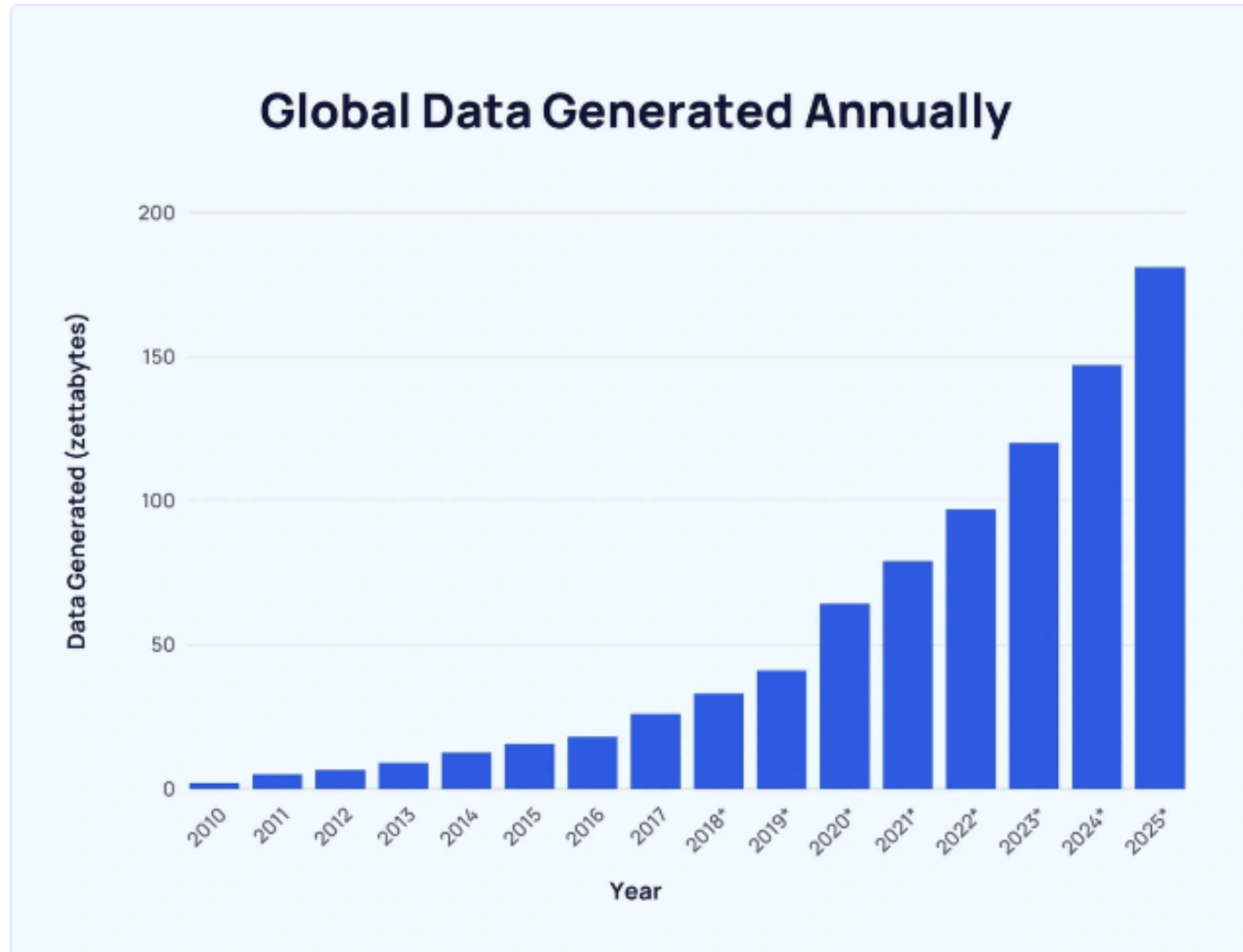
- Senior researcher at the [Department of Political Science](#), University of Zurich
- Principal investigator of the SNSF Ambizione project *[Beyond Traditional Elites: Influencers' Agenda Setting Powers in the Digital Age](#)*
- PhD in Political Science from [Aarhus University](#), Denmark
- Interested in **political attention, public attention, digital platforms, quantitative methods**
- Outside work: snowboarding and flying (I hold a pilot's license)
- Just became a dad: our first daughter was born in August

What about you?

- Name
- Background
- Why did you take the course?
- What do you expect from this course?

What is this course about?

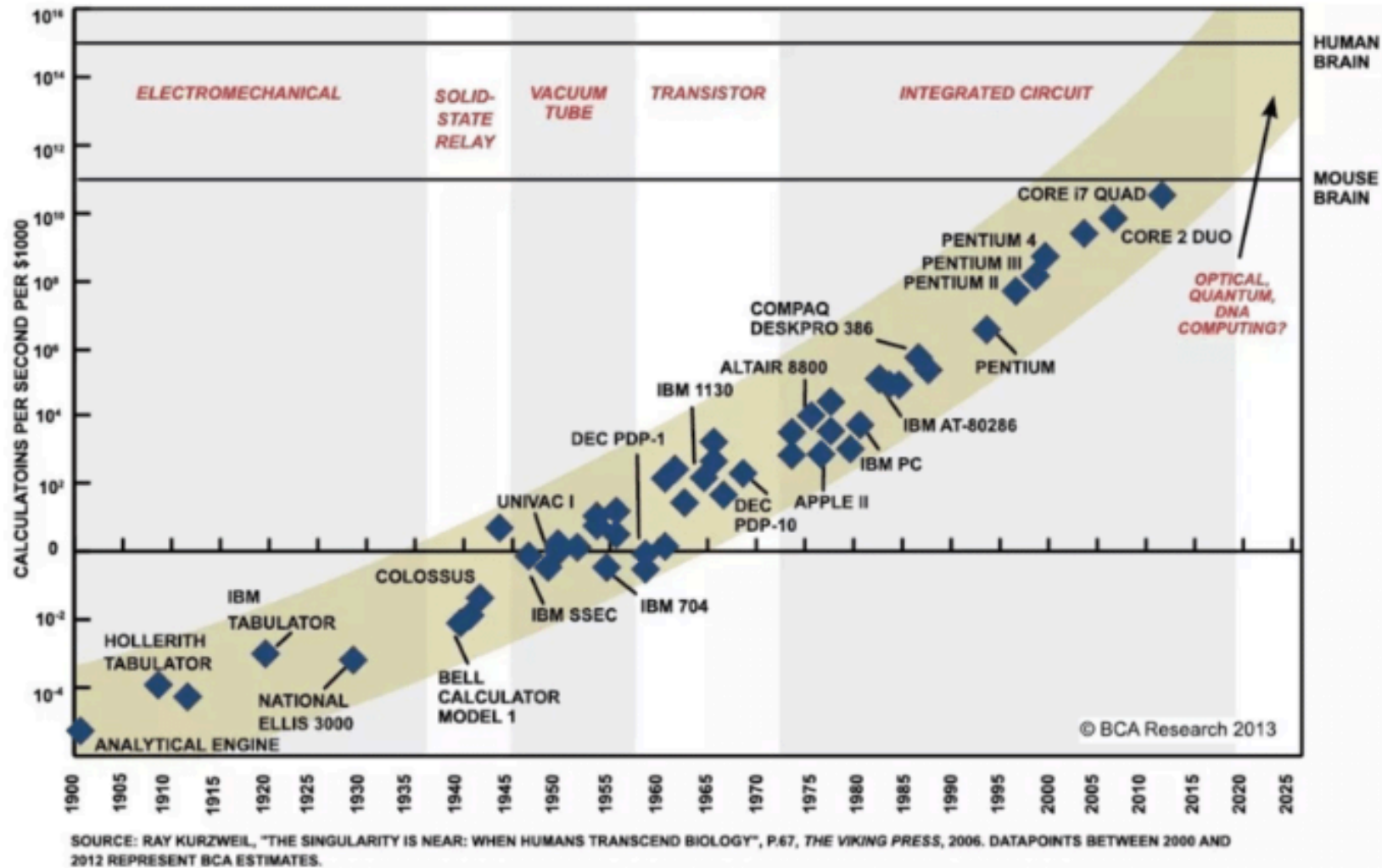
An era of data and computation



[Source here](#)

Masterseminar - Fall Term 2026

An era of data and computation



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Masterseminar - Fall Term 2026

Is it enough with collecting data and having the power to analyze it?

No! Science is all about **design**!

- The goal is to make **inferences** about the world on the basis of empirical information
- To make the right inferences, **methods and rules** are crucial
- This course is about these rules and methods, and how to **apply** them, with a focus on **quantitative research**

In what topics are you interested?



The importance of stupidity in scientific research

Martin A. Schwartz

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Accepted 9 April 2008

Journal of Cell Science 121, 1771 Published by The Company of Biologists 2008

doi:10.1242/jcs.033340

Learning outcomes

1. ...understand the logic of quantitative social science.
2. ...understand and critically assess quantitative research articles.
3. ...identify an appropriate and feasible research design for a given research problem.
4. ...communicate complex concepts effectively to a broad audience.
5. ...perform simple descriptive and inferential statistical analyses.

The course organization

Teaching policy

This is a **seminar**, not a lecture!

Therefore, *active participation* is the most important tool:

- No or only minimal slides
- Basic readings must be read **in advance**
- **Seminar questions** every week
- At least one **presentation** by student

Integration and interaction policy

- No discrimination
- Inclusive language
- No bullying
- **Everyone** should participate in equal terms

Artificial intelligence (AI) policy

- AI tools, like ChatGPT, are allowed
- They should augment, not replace, human work
- Mandatory declaration policy

Evaluation I

1. Attend all the sessions
2. Read the mandatory readings before each session
 - Study the basic reading
3. Participate actively
4. Answer the seminar questions of each session
 - Specific folder in OLAT
 - To be uploaded the day before the class (but not graded!)

Evaluation II

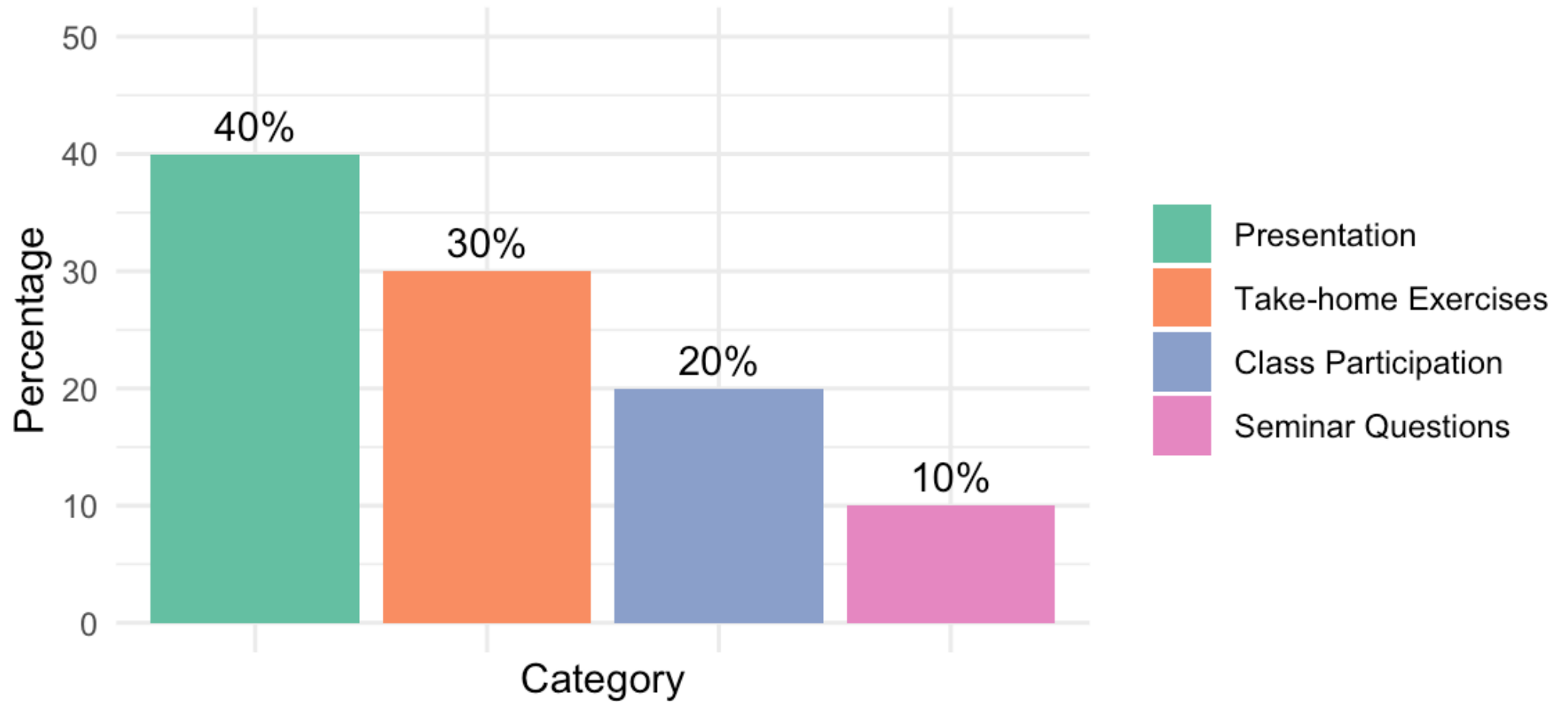
5. Presentation of the applied readings

- More guidelines later
- Organization at the end of this class

6. Complete three take-home exercises

- Second half of the course
- Empirical applications with R
- Two different tracks (beginner and advanced, more on this later)

Grading Policy Breakdown



Masterseminar paper

- **Optional:** as stated in the syllabus, you can choose to write a seminar paper to earn six extra credits
- Between 7000 and 9000 words
- Topic to be agreed with the lecturer
- Paper outline by **December 1st 2026**
 - Research question
 - Academic and societal relevance
 - Theory and hypotheses
 - Approach and structure of the paper
- Submission deadline in Spring, but flexible

Office hours

No office hours!

Send an email to the lecturer (klueser@ipz.uzh.ch) to schedule a meeting one week in advance (**via Zoom**)

Resources (OLAT)

- **AJ:** Angrist, J. D. (2014). Mastering' metrics: The path from cause to effect. Princeton University Press.
- **BCH:** Blair, G., Coppock, A., & Humphreys, M. (2023). [Research design in the social sciences: declaration, diagnosis, and redesign](#). Princeton University Press.
- **BF:** Bueno de Mesquita, E. and Fowler A. (2021). Thinking Clearly with Data: A Guide to Quantitative Reasoning and Analysis. Princeton University Press.
- **GG:** Gerber, A. S., & Green, D. P. (2012). Field experiments: Design, analysis, and interpretation.
- **KKV:** King, G., Keohane, R.O., and S. Verba (1994). Designing Social Inquiry. Princeton: Princeton University Press.
- **KW:** Kellstedt, P. M., and Whitten, G.D. (2013). The Fundamentals of Political Science Research. Third Edition. Cambridge: Cambridge University Press.

Additionally...

- [How to read a scientific article](#)
- [How to critique a scientific article](#)

Schedule I

- Session 1. **Introduction** (17.09.26 / 14:15–16:00 / HS 3)
- Session 2. **The logic of scientific research** (01.10.26 / 14:15–16:00 / 2.A45)
- Session 3. **Theory and research design** (01.10.26 / 16:15–18:00 / 2.A45)
- Session 4. **Data and measurement** (08.10.26 / 14:15–16:00 / 4.B51)

Schedule II

- Session 5. **Descriptive inference** (08.10.26 / 16:15–18:00 / 4.B51)
- Session 6. **Causal inference** (15.10.26 / 14:15–16:00 / 3.B48)
- Session 7. **Predictive inference** (15.10.26 / 16:15–18:00 / 3.B48)
- Session 8. **Experimental studies** (22.10.26 / 14:15–16:00 / 4.B51)
 - Publication take-home exercise I

Schedule III

- Session 9. **Large-N observational studies** (22.10.26 / 16:15–18:00 / 4.B51)
- Session 10. **Small-N observational studies** (29.10.26 / 14:15–16:00 / 3.B47)
 - Deadline take-home exercise I
 - Publication take-home exercise II

Schedule IV

- Session 11. **Statistical testing** (29.10.26 / 16:15–18:00 / 3.B47)
- Session 12. **Introduction to regression I** (05.11.26 / 14:15–16:00 / 4.B51)
 - Deadline take-home exercise II
 - Publication take-home exercise III
- Session 13. **Introduction to regression II** (05.11.26 / 16:15–18:00 / 4.B51)
- Session 14. **Conclusion** (12.11.26 / 14:15–16:00 / 2.A45)
 - Deadline take-home exercise III

Let's dig deeper!

The presentation

- 25 minutes max.
- *Two parts*: paper summary/critique and research proposal
- The summary/critique should focus on the research design and its limitations
- The research proposal should build on these limitations
 - The papers will be good, so **build** on them rather than (try to) destroy them
- After the presentations, there will be a *20 minutes Q&A*

Checklist Part I – Summary (~10 min)

- **Core ideas (1–2 slides, ~3 min):** Summarize the argument and main takeaways.
- **Research design (~4 min):** Present the research question, hypotheses, method, data type, unit of analysis, and key findings.
- **Critique (~3 min):** Identify strengths and main limitations of the research design (measures, validity, scope, etc.). Use clear communication: diagrams, step-by-step reasoning, examples, visual aids.

Checklist Part II – Proposal (~15 min)

- **Address one main limitation:** Focus on *one key issue* and justify why it matters.
- **Propose one solution or extension:** Suggest a feasible new design, replication, different data, or alternative method.
- **Be realistic:** Assume research funds are available, but do not “change the world” (e.g., deciding election dates).
- **Seek guidance if needed:** Contact the lecturer if unsure about preparation.

Additional Information Checklist

- **Evaluation:** graded with the rubric on the next slide.
- **Avoid pitfalls:**
 - Don't overcrowd slides.
 - Avoid superficial content.
 - Manage time (25 min total).
 - Prepare early, rehearse, and refine slides.
 - Practice in front of a friend and ask for feedback.

How your presentation is graded

Each criterion earns **0–5 points**. **15 points** (half the maximum) = 4.0 (pass), **30 points** = 6.0. Pairs receive the same grade. The full conversion table is in the syllabus.

Criterion	Full points when you...
Clarity	Follow the two-part structure, use uncluttered slides with visual aids, and stay within 25 minutes
Understanding	Present the argument, research question, hypotheses, method, data, unit of analysis and findings accurately
Critique quality	Name specific strengths and the main limitations of the research design
Reasoning	Focus on one key limitation, explain why it matters, and derive your proposal from it
Feasibility	Propose a concrete, realistic design that addresses the limitation
Discussion	Answer questions accurately and use the feedback to improve your proposal

Let's organize the calendar!

- Is there a week in which you **absolutely cannot** present? Tell me before the end of today's session.
- I will assign the presentations and share the schedule on OLAT.

Session	Date	Room	Title	Presenter(s)
Session 3	01.10.26 - 16:15	2.A45	Heroes and villains: motivated projection of political identities	
Session 4	08.10.26 - 14:15	4.B51	Measuring democratic backsliding	
Session 5	08.10.26 - 16:15	4.B51	The Manosphere and politics	
Session 6	15.10.26 - 14:15	3.B48	Can exposure to celebrities reduce prejudice? The effect of Mohamed Salah on Islamophobic behaviors and attitudes	
Session 7	15.10.26 - 16:15	3.B48	Predicting results of social science experiments using large language models	
Session 8	22.10.26 - 14:15	4.B51	Beyond Persuasion: Protest's Direct Behavioral Impact on Bystanders	
Session 9	22.10.26 - 16:15	4.B51	Wealth of tongues: Why peripheral regions vote for the radical right in Germany	
Session 10	29.10.26 - 14:15	3.B47	Regulating Uber: The politics of the platform economy in Europe and the United States	
Session 11	29.10.26 - 16:15	3.B47	Without roots: The political consequences of collective economic shocks	
Session 12	05.11.26 - 14:15	4.B51	Allies of the Weak: La Résistance and Jews in the Holocaust	
Session 13	05.11.26 - 16:15	4.B51	Plata y plomo: How higher wages expose politicians to criminal violence	

Three take-home assignments

- Autonomous learning of **R** and **RStudio**
- Materials will be provided by the lecturer
- ***Goal:*** gain hands-on experience with class topics
- Two voluntary tracks depending on your familiarity with R:
 - **Beginner** – installation, getting started with RStudio, comment (maybe perform) basic statistical analyses
 - **Advanced** – everything in Beginner + additional statistical analyses

Which track fits you? Self-assessment

Rate each statement: 0 = not at all · 1 = with help · 2 = confidently on my own

R and RStudio

1. I have R and RStudio installed and can open and run an R script.
2. I can load a dataset (e.g. `.csv` or `.rds`) into R and inspect its variables.
3. I can clean and transform variables (recode, handle missing values, create categories).
4. I can plot distributions and relationships between variables (e.g. with `ggplot2`).
5. I can write a Quarto or R Markdown document and render it to PDF.

Self-assessment (continued)

Statistics

6. I can compute and interpret descriptive statistics (means, standard deviations, cross-tabulations).
7. I can run and interpret a difference-in-means test (e.g. a t-test).
8. I can estimate an OLS regression in R and interpret its coefficients, including interaction terms.

Add up your points (0–16)

- **0–9 points:** beginner track recommended
- **10–16 points:** advanced track recommended
- The score is a guide. You choose your track, and you are evaluated against the expectations of the track you choose.

Please send me an email with your decision next week!



Conclusion

Next session (01.10.26 - 14:15)

The logic of scientific research

Presentation of ongoing research by the lecturer:

- **Klüser, S., Klüser, K. J., & Hoes, E. (2026).** AI-Generated Explicit Deepfakes Damage Politicians' Perceived Leadership Competence, Trustworthiness and Electoral Prospects. *Revised and resubmitted, Communications Psychology*

Do not forget to register at UniPortal!

UNIVERSITÄT
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WICHTIG: ANMELDUNG ZUR LEHRVERANSTALTUNG

Um Credits für diese Veranstaltung zu erwerben, ist eine Anmeldung über das **UniPortal** innerhalb des Anmeldezeitraums **zwingend erforderlich**.

Die Einschreibung auf OLAT genügt nicht.

**Anmeldezeitraum UniPortal:
31. August – 25. September 2026**

IMPORTANT: COURSE REGISTRATION

In order to acquire credits for this course, registration via **UniPortal** within the registration period is **mandatory**.
Enrolment on OLAT is not sufficient.

**Registration period UniPortal:
31 August – 25 September 2026**



UniPortal



Thanks! 🍌

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